# HARSHANA KARUNARATHNE

Software Engineer | Java, Spring Boot, React
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## PROFESSIONAL SUMMARY

Full-stack Software Engineer with 2+ years of experience designing and delivering enterprise-grade applications with Java, Spring Boot and React. Builds secure systems end to end, from a modular Spring Boot backend and versioned REST APIs to real-time WebSocket updates, Electron hardware integration and AI voice ordering. Experienced in banking and payment systems, multi-branch platforms, role-based security, audit compliance, event-driven processing with Apache Kafka and modernizing legacy Java applications. Graduate in Statistics and Computer Science.

## TECHNICAL SKILLS

**Languages:** Java (8, 17, 21), TypeScript, JavaScript, Python, SQL

**Backend:** Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Spring AI, Hibernate, REST APIs, Modular Monolith Architecture

**Frontend:** React, Tailwind CSS, TanStack Query, Electron, Thymeleaf

**Messaging and Real-Time:** WebSockets, Webhooks, Apache Kafka, MQTT

**Databases:** MySQL, MSSQL, PostgreSQL, MongoDB, Firebase, Flyway

**Cloud and DevOps:** AWS, Docker, Kubernetes, CI/CD (GitHub Actions), Git, GitHub, GitLab

**Security and Testing:** JWT, Passkeys (WebAuthn), Role-Based Access Control (RBAC), Audit Logging, VAPT Remediation, JUnit, Vitest

**AI:** Google Gemini, Gemini Live API, LLM Function Calling

## WORK EXPERIENCE

**Software Engineer, TS Technologies (Pvt) Ltd** **Sep 2024 – Present**

*Java 21, Spring Boot, React, TypeScript, MySQL, Flyway, Electron, Docker*

- Full-stack engineer on the company's restaurant commerce product line – a multi-branch POS and an AI voice-ordering kiosk – taking features from design through the Spring Boot backend, React front ends and Electron hardware integration to release.
- Designed the integration between the POS and kiosk products: two independently deployed Spring Boot backends, each with its own MySQL database, connected only through a versioned REST and webhook contract.
- Built reusable React components and Spring Boot modules with Flyway migrations and automated tests (JUnit, Vitest) run in GitHub Actions CI, and contributed to the company's CRM platform.

**Software Engineer Intern, Epic Lanka (Pvt) Ltd** **Feb 2024 – Sep 2024**

*Java 17, Spring Boot, Spring MVC, Apache Kafka, MSSQL, Docker, AWS*

- Modernized a centralized terminal management system supporting 5,000+ POS devices by migrating it from Struts (Java 8) to Java 17 with Spring Boot, improving scalability, performance and maintainability.
- Designed and delivered payment processing and billing microservices with Spring Boot and Apache Kafka, automating event-driven transaction workflows.
- Built RESTful APIs and server-rendered views with Spring MVC, Thymeleaf and MSSQL for secure enterprise data handling.
- Enforced enterprise security standards by resolving VAPT findings and implementing audit logging with 100% traceability across terminals.
- Improved the stability of production backend services in a banking environment through performance tuning and debugging.

## PROJECTS

**TS POS and AI Voice Kiosk – Restaurant Commerce Platform** **TS Technologies**

*Java 21*, *Spring Boot*, *MySQL*, *Flyway*, *React*, *TypeScript*, *Tailwind CSS*, *Electron*, *Spring AI*, *Google Gemini Live API*, *WebSockets*, *Docker*

- Built the POS backend as a modular Spring Boot application – 30+ domain modules (orders, payments, pricing, inventory, kitchen, reporting) in a layered architecture on one MySQL database, with each optional capability behind a server-enforced module toggle.
- Built the POS suite – sales terminal, Kitchen Display System, admin portal and QR guest menu – with a central pricing engine for taxes, discounts and promotions, and idempotent order and payment APIs that prevent duplicate charges.
- Built AI voice ordering on the kiosk with Spring AI and the Gemini Live API, reducing average order time by 40%, and a recommendation engine that raised average order value by 25%.
- Delivered real-time updates to terminals, kitchen displays and kiosks over WebSockets, offline sales that sync on reconnect, Electron-driven receipt printers and cash drawers, and multi-branch RBAC with JWT, audit logging and branch-scoped data access.

**Customer Relationship Management System** **TS Technologies**

- Developed CRM modules for customer management, lead tracking and sales workflow automation.
- Implemented AI-driven lead prioritization to reduce manual lead analysis.
- Optimized database queries and backend services to speed up page loads and reporting, and automated routine business processes.

**ATEM Hub – Terminal Management System for BML** **Epic Lanka**

- Revamped a centralized platform for managing POS applications and devices, migrating it from Struts (Java 8) to Java 17 with Spring Boot and Spring MVC, with REST APIs, Thymeleaf server-side rendering and MSSQL.

**Automatic Bill Settlement** **Epic Lanka**

- Built Payment Processor and Billing Mediator microservices for a next-generation payment system using Java 17, Spring Boot, React and Apache Kafka, deployed in Docker containers on AWS with cloud scheduling, load balancing and VAPT-compliant security.

## EDUCATION

**B.Sc. in Physical Science (Statistics and Computer Science)** **2022 – 2025**

University of Kelaniya | Dean's List, awarded every academic year

**G.C.E. Advanced Level, Physical Science Stream** **2017 – 2020**

Royal College, Colombo | Combined Mathematics A, Chemistry B, Physics C

## REFERENCES

Dr. Basitha, Senior Lecturer, University of Kelaniya | basitha@kln.ac.lk

Mr. Muditha Hettiarachchi, Director, Success Delivery Operations, IFS | muditha.hettiarachchi@ifs.com